Method of expressing concentration of solution

61. (d) Molar concentration $[H_2] = \frac{Mole}{Vinlitre} = \frac{20/2}{5} = 2.$

62. (a) Amount of $AgNO_3$ added in 60 ml of solution $= 60 \times 0.03 = 1.8\ g$

63. (a) $N = \frac{w}{E \times V(l)} \Rightarrow 0.1 = \frac{w}{100 \times 0.1} \Rightarrow w = 1gm$

64. (b) $N = \frac{w}{E \times V(l)} \Rightarrow 0.1 = \frac{w}{40 \times 0.25} \Rightarrow w = 1gm$

65. (a) $20 \times 0.4 = 40 \times N$ or $N = 0.2$ or $M = \frac{0.2}{2} = 0.1M.$

69. (a) $M = \frac{w \times 1000}{m.wt. \times \text{Volume in } ml.} = \frac{10.6 \times 1000}{106 \times 500} = 0.2M.$

72. (c) M.eq. of HCl = M.eq. of $CaCO_3$

$N \times 50 = \frac{1}{50} \times 1000$; $N = \frac{1 \times 1000}{50 \times 50} = 0.4N$

73. (c) molality $= \frac{18}{180} = 0.1molal.$

74. (b) Molarity of $H_2SO_4 = 0.5$

Normality of $H_2SO_4(N_1) = 0.5 \times 2 = 1$

$N_1V_1 = N_2V_2$

$1 \times 1 = N_2 \times 10$ or $N_2 = \frac{1}{10} = 0.1N.$

76. (c) The density of solution $= 1.8gm/ml$

Weight of one litre of solution $= 1800gm$

$\therefore$Weight of H_2SO_4 in the solution$= \frac{1800\times90}{100}$=162$gm$

$\therefore$Weight of solvent $= 1800 - 1620 = 180gm$

$\therefore$Molality $= \frac{1620}{98} \times \frac{100}{180} = 9.18$

77. (a) Suppose the total volume of water = x

$\therefore 100cm^3 \times 0.5N = x \times 0.1N$

$\therefore x = \frac{100\times0.5}{0.1} = 500cm^3$

Therefore the volume of water added

$= \text{Totalvolume}-100\text{cm}^3 = 500 - 100 = 400cm^3$.

78. (b) $M_1V_1 = M_2V_2$, $M_2 = \frac{0.25\times25}{500} = 0.0125$.

79. (c) % by $wt. = \frac{wt.\text{of the solute } (g)}{wt.\text{ of the solution } g} \times 100$

$= \frac{10}{90+10} \times 100 = 10$

80. (b) Molality $= \frac{w}{m\times W} \times 1000 = \frac{18\times1000}{180\times250} = 0.4m$